

Elementary Problems Counting

Version 1.0

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Basic Work

1. In how many ways can we pair up 10 people to form 5 pairs?
2. Let S be a set consisting of n elements. How many nonempty subsets of S are there?
3. In how many ways can we partition the positive integers from 1 to n into 3 disjoint nonempty sets so that no two consecutive integers belong to the same set?
4. Let n be a positive integer. In how many ways can we put an integer from 1 to n in each of the cells of an $n \times n$ table so that the sum of numbers in each row and each column is divisible by n ?
5. Let $n \geq 3$ be a positive integer. Find the number of 3-element subset of $\{1, 2, \dots, n\}$ so that the three elements form an arithmetic progression.
6. Let $n \geq 3$ be a positive integer. How many 3-element subsets of $\{1, 2, \dots, n\}$ are there whose sums of elements are multiples of 3?
7. Let $n \geq 3$ be a positive integer. How many ways are there can we put 2 identical kings on an $n \times n$ chessboard so that they do not threaten each other? The kings threaten each other if they are put on two cells sharing a common vertex or a common edge.
8. (IMO shortlist 1969) Find the number of five-digits numbers with the following properties: there are two pairs of digits such that digits from each pair are equal and are next to each other, digits from different pairs are different, and the remaining digit (which does not belong to any of the pairs) is different from the other digits.
9. (APMO 2001 Problem 2) Find the largest positive integer N so that the number of integers in the set $\{1, 2, \dots, N\}$ which are divisible by 3 is equal to the number of integers which are divisible by 5 or 7 (or both).
10. How many positive integers not exceeding 1000 are there which are not divisible by 4, 5 or 6?
11. 5 people sit in a row. In how many ways can the people be rearranged so that none of them takes the original seat?
12. Given a regular polygon with $n \geq 3$ sides, how many isosceles triangles (including equilateral triangles) can be formed using the vertices of this polygon?
13. (CWMO 2007 Problem 1) It is known that $T = \{1, 2, 3, 4, 5, 6, 7, 8\}$. For any $A \subseteq T$, $A \neq \emptyset$, define $S(A)$ to be the sum of elements in A . How many nonempty subsets A does T have such that $S(A)$ is a multiple of 3 but not a multiple of 5?

14. (Costa Rica Final Round 2003 Problem 5) Each of the squares of an 8×8 board can be coloured white or black. Find the number of colourings of the board such that every 2×2 square contains exactly 2 black squares and 2 white squares.

Permutation and Combination

15. How many ways are there can we use the digits 0, 0, 1, 1, 2, 2, 3, 3 to form an 8-digit number?
16. How many ways are there can we put the digits 0, 0, 1, 1, 2, 2 in a cycle? (Two ways are regarded as the same if one can be obtained from the other by a rotation.)
17. How many ways are there can we rearrange the integers from 1 to 9 to form a sequence so that the sequence is first increasing and then decreasing? (The sequence cannot be entirely increasing or entirely decreasing.)
18. How many ways are there can we put 10 chess pieces on an 8×8 board so that we can select a row and a column which together contain all chess pieces?
19. How many ways are there can we choose 3 distinct integers from 1 to 99999 such that the 3 integers together consist of exactly 2 distinct digits?
20. Let $k \leq n$ be positive integers. In how many ways can we put k identical rooks on an $n \times n$ chessboard so that no two rooks threaten each other? Two rooks threaten each other if they lie in the same row or the same column.
21. Let $n \geq 3$ be an integer and let $S = \{1, 2, \dots, n\}$.
- (a) How many functions $f : S \rightarrow S$ are there such that $f(1) < f(2) < f(3)$?
 - (b) How many bijective functions $g : S \rightarrow S$ are there such that $g(1) < g(2) < g(3)$?
22. Let n and k be positive integers with $k \leq n$. How many ways are there to partition the set $\{1, 2, \dots, n\}$ into k nonempty subsets? Here, the subsets are indistinguishable. So swapping two subsets give the same partition.

Recurrence Sequence

23. In how many ways can we climb a ladder of n steps if we can only move 1 step or 2 steps upwards each time?
24. How many bijective functions f of the set $\{1, 2, \dots, n\}$ are there such that $|f(k) - k| \leq 1$ for each $k = 1, 2, \dots, n$?

25. We start from the vertex A of a regular tetrahedron $ABCD$. Each step we can move to any other vertex. How many routes of n steps are there which end at A ?
26. How many n -digit numbers are there such that each digit is one of $1, 2, 3$, and no three consecutive digits are the same?
27. How many n -digit numbers are there such that each digit is one of $1, 2, 3$, and no three consecutive digits are pairwise distinct?
28. (IMO shortlist 1987) How many words with n digits can be formed from the alphabet $\{0, 1, 2, 3, 4\}$, if neighbouring digits must differ by exactly one?
29. (IMO 1979 Problem 6) Let A and E be opposite vertices of a regular octagon. A frog starts jumping at vertex A . From any vertex of the octagon except E , it may jump to either of the two adjacent vertices. When it reaches vertex E , the frog stops and stays there. Let a_n be the number of distinct paths of exactly n jumps ending at E . Prove that $a_{2n-1} = 0$,

$$a_{2n} = \frac{1}{\sqrt{2}}(x^{n-1} - y^{n-1}), \quad n = 1, 2, 3, \dots,$$

where $x = 2 + \sqrt{2}$ and $y = 2 - \sqrt{2}$.

Note. A path of n jumps is a sequence of vertices $(P_0, \dots, P_n)$ such that

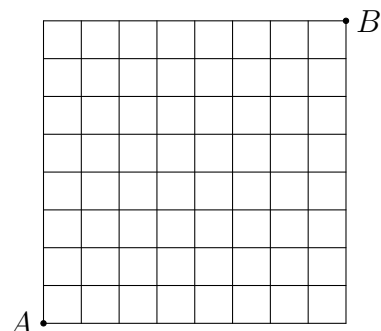
- (i) $P_0 = A, P_n = E$;
 - (ii) for every $i, 0 \leq i \leq n-1$, P_i is distinct from E ;
 - (iii) for every $i, 0 \leq i \leq n-1$, P_i and P_{i+1} are adjacent.
30. Let m and n be positive integers. How many tuples $(a_1, a_2, \dots, a_m)$ of nonnegative integers are there such that $0 \leq a_j \leq n$ for each j and $n \mid a_1 + a_2 + \dots + a_m$?
31. (Turkey TST 2014 Problem 1) Find the number of permutations $(a_1, a_2, \dots, a_{2014})$ of $(1, 2, \dots, 2014)$ such that, for all $1 \leq i < j \leq 2014$, $i + a_i \leq j + a_j$.
32. (IMO shortlist 2008 C2) For every positive integer n determine the number of permutations $(a_1, a_2, \dots, a_n)$ of the set $\{1, 2, \dots, n\}$ with the following property:

$$2(a_1 + a_2 + \dots + a_k) \text{ is divisible by } k \text{ for } k = 1, 2, \dots, n.$$

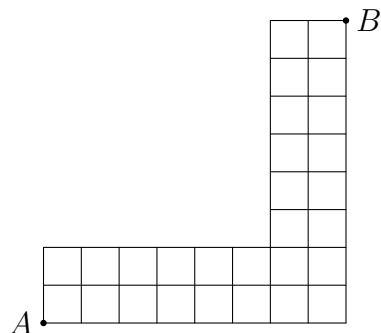
33. (Vietnam MO 2009 Problem 5) Let $S = \{1, 2, 3, \dots, 2n\}$ ($n \in \mathbb{Z}^+$). Determine the number of subsets T of S such that there are no 2 elements a, b in T such that $|a - b|$ is 1 or n .

Bijection

34. How many tuples of nonnegative integer solutions to the equation $a + b + c + d = 30$ are there?
35. How many tuples of positive integer solutions to the equation $a + b + c + d = 30$ are there?
36. How many tuples of integer solutions to the equation $a + b + c + d = 30$ are there such that $3 \leq a, b, c, d \leq 9$?
37. How many tuples of nonnegative integers (a, b, c, d) are there such that $a + b + c + d \leq 30$?
38. (Putnam 2005 B4) For positive integers m and n , let $f(m, n)$ denote the number of n -tuples $(x_1, x_2, \dots, x_n)$ of integers such that $|x_1| + |x_2| + \dots + |x_n| \leq m$. Show that $f(m, n) = f(n, m)$.
39. In how many ways can we walk from A to B if we can only move upwards or rightwards in each step?



40. In how many ways can we walk from A to B if we can only move upwards or rightwards in each step?



41. In how many ways can we walk from $(0, 0)$ to (n, n) in the coordinate plane if we can only move upwards or rightwards by 1 unit in each step and we cannot move to a point on the line $y = x + 1$?
42. How many n -digit numbers are there which consist only of the digits 1, 2, 3, 4 and are multiples of 3? (It is not necessary that all digits are used.)
43. (IMO shortlist 1980) Prove that $\sum \frac{1}{i_1 i_2 \cdots i_k} = n$ where the sum is taken over all nonempty subsets $\{i_1, i_2, \dots, i_k\}$ of $\{1, 2, \dots, n\}$. (The k is not fixed, so we are summing over all the $2^n - 1$ possible nonempty subsets.)
44. (IMO shortlist 1987) Find the number of partitions of the set $\{1, 2, \dots, n\}$ into three subsets A_1, A_2, A_3 , some of which may be empty, such that the following conditions are satisfied:
- (i) After the elements of every subset have been put in ascending order, every two consecutive elements of any subset have different parities.
 - (ii) If A_1, A_2, A_3 are all nonempty, then in exactly one of them the minimal number is even.

(Remark: We may regard A_1, A_2, A_3 as different subsets. So swapping two subsets give a new partition.)

Record

v1.0: problems 1–44